

V-NET LIGHT - PARAMETER-EFFICIENT 3-D CONVOLUTIONAL NEURAL NETWORK FOR PROSTATE MRI SEGMENTATION

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ABSTRACT

Prostate MRI segmentation has become an important tool for quantitative estimation of the gland volume during diagnostic imaging. It is also a critical step in the fusion between MRI and transrectal ultrasound (TRUS) for fusion guided biopsy or therapy. 3-D neural networks have demonstrated strong potential for this task, but require substantial computational resources due to their large number of parameters. In this work, we focus on the efficiency of the segmentation network in terms of speed and memory requirements. Specifically, we aim at reaching state-of-the-art results with smaller networks, involving significantly fewer parameters, thus making the network easier to train and operate. A novel 3-D network architecture, called *V-net Light* (*VnL*) is proposed, based on an efficient 3-D Module called *3-D Light*, that minimizes the number of network parameters while maintaining state-of-the-art segmentation results. The proposed method is validated on the PROMISE12 challenge data [1]. The proposed *VnL* has only 9.1% of V-net's parameters, 3.2% of its floating point operations (FLOPs) and uses only 9.1% of hard-disk storage compared to V-net, yet V-net and *VnL* has comparable accuracy.

Index Terms— Prostate segmentation, Deep learning, Neural network, Small model, Fast training

1. INTRODUCTION

Prostate cancer is the second most common cancer among men, but with early detection it can often be treated successfully. Multi-parametric Magnetic Resonances Imaging (MRI) is an effective way to visualize the prostate gland. Segmenting the prostate within an MRI scan provides valuable assessment of the prostate volume and mass, used in primary diagnosis and follow-up. Moreover, fast and accurate prostate segmentation is a must in the registration between MRI and transrectal ultrasound (TRUS) during fusion guided biopsy or therapy [2, 3].

In recent years, several works have been published on prostate MRI segmentation using 2-D or 3-D deep neural networks. Among the proposed 2-D approaches, the U-net architecture [4] has been very successful. It is based on a U-shaped CNN structure comprising a left branch, the contracting path, and a right branch, the expanding path. The contracting path consists of convolution layers, each followed by max pooling that progressively reduces the map size. In the consecutive expanding path, skip connections followed by deconvolutions up-sample the segmented map back to the size of the input image. In [5], a 2-D encoder-decoder approach, that

uses dense residual connections between layers, was proposed. The idea was inspired by DenseNet [6], in which the dense connections were initially proposed. Considering the 3-D nature of anatomy, 3-D network architectures may prove advantageous. Under this assumption, 3-D U-net [7] was proposed as a direct 3-D extension of 2-D U-net, using 3-D kernels. V-net [8] went further by applying, besides 3-D convolutions, the minimization of a Dice loss during the training phase, and reached state-of-the-art results for prostate segmentation. Recently, further improvements of the 3-D architectures were proposed to deal with the vanishing gradients problem: Wang [9] used a deeply supervised 3-D FCN, while Yu [10] proposed mixed long and short residual connections.

While 3-D architectures can better represent the prostate, their implementation is more challenging in terms of computing resources given the large number of parameters and FLOPs they involve. Consider, TRUS-MRI fusion biopsy performed at a private practice. It would greatly benefit from an automatic prostate segmentation running on limited power hardware embedded in the TRUS imaging station, avoiding expensive workstations or remote computing clouds that can deal with tens of millions of parameters.

In this paper, we propose a novel 3-D architecture that minimizes the size of the network, i.e. the number of parameters in the model. The proposed approach replaces regular 3-D convolutions of the V-net architecture with novel *3-D Light modules* that dramatically reduce the number of parameters without significantly affecting the segmentation results.

2. METHOD

The proposed macro network structure (Fig. 1) and the novel *3-D Light Module* (Fig. 2) are presented in detail in the following subsections.

2.1. Macro Structure

The macro structure (Fig. 1) of the suggested 3-D network is based on [4, 8] which proposed building a network with two paths: the compression path, or *encoder*, and the decompression path, or *decoder*. Both paths implement convolutions, with feature map resolution changing from layer to layer. Since our goal is to reduce the number of parameters, we insert a pooling layer between the encoder stages instead of convolutions with stride 2. At the decoder side, we perform *Transposed convolution* [11] (deconvolution) to learn the best up-sampling parameters. The output of every encoder stage is fed to the next stage and also added as a skip connection input to

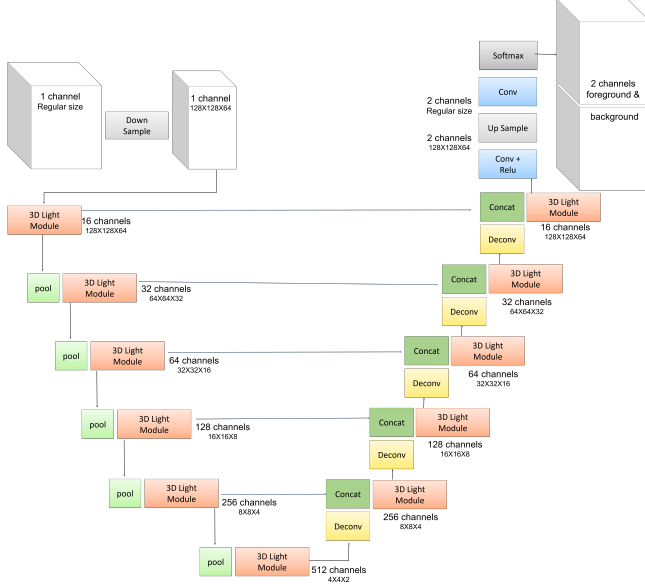


Fig. 1. Macro structure network

the same resolution stages in the decoder side, to enforce residual learning [12].

In each encoder block, we progressively decrease the feature map resolution by a factor of 2 while increasing the number of channels by the same factor. Conversely, in the decoder blocks, the feature map resolution increases by 2 whereas the number of channels decreases by the same factor. Each stage of the encoder and decoder comprises a novel *3-D Light Module* described in the following subsection.

2.2. 3-D Light Module

The proposed *3-D Light Module* is a parameter-efficient 3-D convolutional block inspired by the 2-D *Fire Module* introduced in [13]. The innovative structure of the *3-D Light Module* is shown in Fig. 2. Unlike the *Fire Module*, the *3-D Light Module* implements 3D and group convolutions [14, 15]. It begins with a $1 \times 1 \times 1$ convolution layer, that squeezes the number of channels. In Fig. 2, the number of channels is reduced from 16 to 4. Then, parallel $1 \times 1 \times 1$ and $3 \times 3 \times 3$ convolution blocks expand back the number of channels (to 16 channels in Fig. 2), which are finally concatenated. These parallel convolution blocks are composed of a regular convolutions followed by a group convolution. As shown in Fig. 3 for g groups, C_{out}/g 3-D kernels are used for each group, each group operating on a different subset of C_{in}/g input channels (where C_{in} and C_{out} are the number of input and output channels respectively). For example, for $g = 2$, the kernels in each group are of size $3 \times 3 \times 3 \times C_{in}/2$ or $1 \times 1 \times 1 \times C_{in}/2$, while each group contains $C_{out}/2$ kernels. The outputs are concatenated to C_{out} channels.

As shown in Table 1 for different convolution group sizes (1,2 and 8 group sizes are denoted by 'VnL', 'VnL g2', 'VnL g8'), the proposed network, based on the *3-D Light Module*, reduces the number of parameters by 88%-92% in comparison to V-net.

Instance Normalization [16] and *Relu* are applied after each convolutional layer. Training and testing using Instance Normalization addresses the issue of specific instance contrast that is of concern with MRI data.

Table 1. Comparison of the number of parameters between standard V-net to VnL.

Method	Number of parameters 16 channels input and 32 output stage	Total model parameters (millions)
V-net	64,000	66.0
VnL	9,092	7.6
VnL g2	5,508	6.0
VnL g8	2,820	4.8

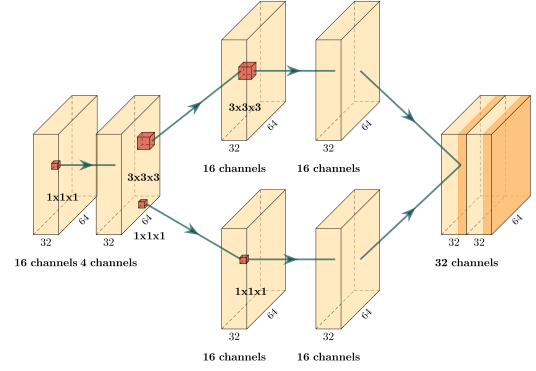


Fig. 2. 3-D Light Module:
example with input $64 \times 64 \times 32 \times 16$ and output $64 \times 64 \times 32 \times 32$ ¹.

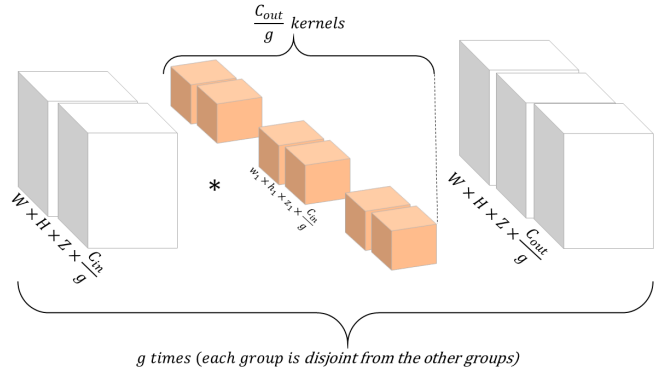


Fig. 3. 3-D group convolution with group size g where the input size is $W \times H \times Z \times C_{in}$, output size is $W \times H \times Z \times C_{out}$. Each group has C_{out}/g kernels, and each kernel is $w_1 \times h_1 \times z_1 \times C_{in}/g$

2.3. Dice Loss Function

The proposed network architecture is trained using the Dice Similarity Coefficient (DSC) [8, 17] as loss function. This loss can be written as:

$$D_{loss} = 1 - \frac{2 \sum_{n=0}^N p_n^2 \cdot g_n^2}{\sum_{n=0}^N p_n^2 + \sum_{n=0}^N g_n^2} \quad (1)$$

where p_n and g_n are the prediction and the ground-truth for voxel n , respectively, and N is the number of voxels in each volume.

3. EXPERIMENTS

The architecture described in the previous section was trained on 50 T2 MRI scans from the PROMISE12 data set [1]. The data was acquired at several institutions, with different equipment and acquisition protocols. For comparison, we also implemented the original V-net architecture². The performance of the proposed method was assessed using the Dice Similarity Coefficient (DSC), the 95th percentile of the asymmetric Hausdorff Distance (95HD), and the Average Symmetric Surface Distance (ASSD) [1] in a 10-Fold-Cross-Validation framework. Each fold was trained on 45 T2 MRIs scans and was validated on 5 different scans i.e., each scan was used for validation exactly once. The proposed architecture was trained for 300 epochs with $batch_size = 2$, using an Nvidia GeForce GTX 1070 GPU with 8G RAM. The network parameters were optimized by an Adam algorithm with learning rate of 0.001.

4. RESULTS

In Fig. 4, the DSC obtained for the proposed method (blue, orange, gray) and V-net (yellow) are compared over the 50 cases cross-validated dataset. Different group sizes are considered for the group convolution: 8, 2 and 1, denoted by $VnL\ g8$ (blue), $VnL\ g2$ (orange), and VnL (gray), respectively. Examples of the segmentation results are shown in Fig. 5.

The results are further summarized in Table 2. In terms of DSC, the proposed VnL and $VnL\ g8$ are comparable to V-net, whereas their FLOPs and memory requirements (last two columns in Table 2) are 3.7%-3.0% and 11.4%-7.3% of V-net's own, respectively. Moreover, by using a 'Paired Sample T-Test' on the DSC, we observed that $VnL\ g2$ outperforms V-net ($p < 0.02$) with just 3.2% of V-net's FLOPs and 9.1% of its memory requirements.

Table 2. Results comparison between VnL and standard V-net. Dice Similarity Coefficient (DSC), 95% Hausdorff Distance (95HD), Average Symmetric Surface Distance (ASSD), Floating Point Operations (FLOPs) and Size of weights file (Size).

Method	DSC	95HD [mm]	ASSD [mm]	FLOPs [GB]	Size [MB]
<i>V-net</i>	0.85 ± 0.072	2.202 ± 1.25	0.311 ± 0.11	867.8	793.1
<i>VnL</i>	0.85 ± 0.094	2.536 ± 1.50	0.309 ± 0.14	32.4	90.6
<i>VnL g2</i>	0.86 ± 0.065	2.270 ± 1.10	0.321 ± 0.14	28.5	72.2
<i>VnL g8</i>	0.85 ± 0.073	2.736 ± 1.22	0.335 ± 0.13	25.5	58.5

5. CONCLUSIONS

In this paper, we proposed a novel 3-D architecture that is particularly efficient in terms of parameters, memory and FLOPs for 3-D segmentation tasks. The proposed method is successfully applied to the 3-D segmentation of prostate MRI. Quantitative segmentation results similar to those of state-of-the-art V-net are obtained while requiring 90% less learning parameters, 90% less hard-disk storage and just 3.3% of the FLOPs.

¹The figure was created using: <https://github.com/HarisIqbal88/PlotNeuralNet>

²V-net model was created with help from:

<https://github.com/mattmacy/vnet.pytorch>

The *V-net Light* architecture presented in this paper is a promising way to facilitate computationally-heavy 3-D segmentation tasks on limited-resource medical imaging devices, such as the TRUS-MRI fusion biopsy system.

6. REFERENCES

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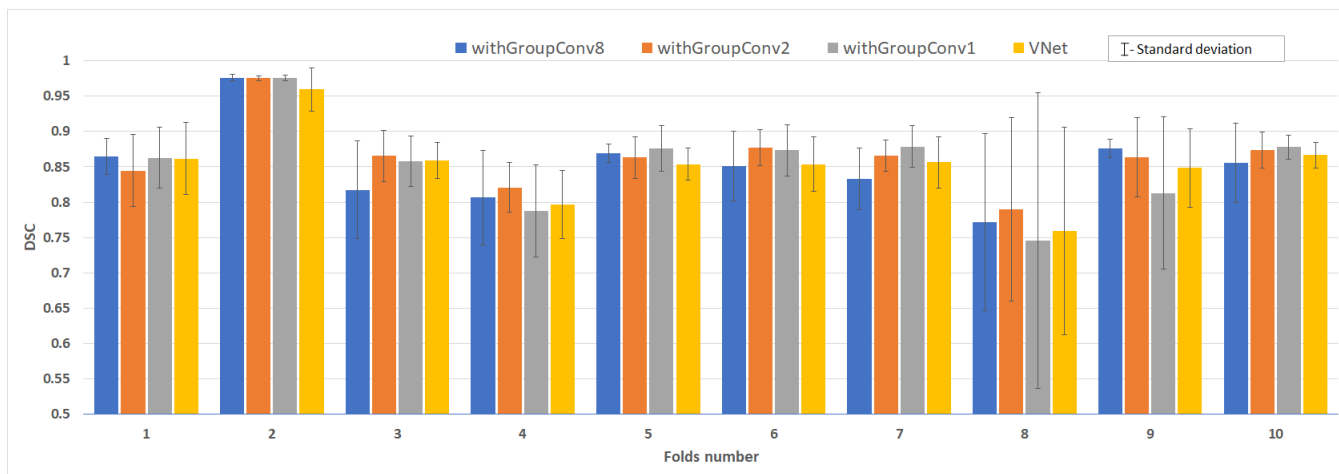


Fig. 4. Average Dice Similarity Coefficient obtained for the proposed method and V-net over the 10-Fold-Cross-Validation sets. Different group convolution sizes are considered for the proposed method: 8, 2 and 1. we observed that the different *V-net Light* versions generally demonstrate similar accuracy to the regular V-net

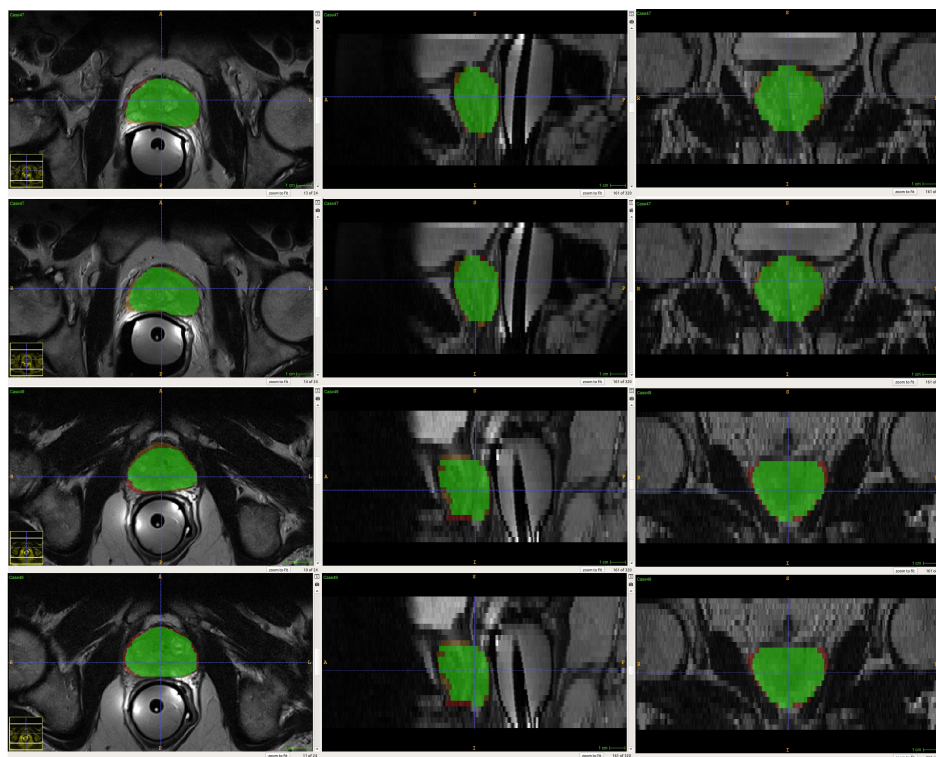


Fig. 5. Results from VnL on axial, sagittal and coronal prostate MRI views (left to right). Top two rows are two sample slices for one case and last rows are for another case. False negative (red), false positive (orange), and true positive (green) voxels are marked on the views.

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